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Electric clipper having integrated hair cutting and suction

Abstract

An electric clipper includes a face cover, an intermediate cover, a bottom cover and an electric cutting assembly. The face cover and the bottom cover are buckled on two sides of the intermediate cover respectively. A first accommodating space is formed between the intermediate cover and the bottom cover. The electric cutting assembly is fixedly arranged in the first accommodating space. A suction channel is formed between the face cover and the intermediate cover. The air inlet end of the suction channel is matched with the electric cutting assembly. An integration of hair cutting and suction is implemented.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to hairdressing devices and more particularly to an electric clipper having integrated hair cutting and suction.

2. Description of Related Art

(2) When people use traditional electric clippers for hairdressing, they usually tie a piece of cloth around their bodies and collect clipped hairs through external suction devices, preventing clipped hairs from falling onto their bodies to a certain extent; however, despite of wrapping a piece of cloth, when males with short hair get their hair cut, clipped hair may still fall into their coats from

the collars, leading to itching on their bodies. When people use electric clippers to shear animals such as cats, dogs and sheep, they have to lay a piece of cloth on the ground and put the animals on the cloth for shearing, and then collect the clipped furs falling on the ground through external suction devices; however, the furs are still easy to fly everywhere in the falling process, making it difficult to collect them. In this context, the applicant devotes to developing a hair cutting and suction integrated electric clipper which is more favorable for collecting clipped hair or furs through the external suction devices.

SUMMARY OF THE INVENTION

(3) The invention aims to realize more efficient collection of clipped hair or furs through external suction devices.

(4) In order to achieve the above objective, the invention adopts the technical solution as follows:

(5) A hair cutting and suction integrated electric clipper includes a face cover, an intermediate cover, a bottom cover and an electric cutting assembly, wherein the face cover and the bottom cover are buckled on two sides of the intermediate cover respectively, a first accommodating space is formed between the intermediate cover and the bottom cover, the electric cutting assembly is fixedly arranged in the first accommodating space, a suction channel is formed between the face cover and the intermediate cover, and the air inlet end of the suction channel is matched with the electric cutting assembly.

(6) Further, the hair cutting and suction integrated electric clipper includes an external suction connector, wherein two ends of the external suction connector penetrate through the external suction connector, and one end of the external suction connector is buckled at the air outlet end of the suction channel.

(7) Further, the hair cutting and suction integrated electric clipper includes an air blower and a baffle plate, an air inlet is formed in the center of the intermediate cover, an air outlet is formed in the bottom cover; the air blower is fixedly arranged in the first accommodating space, the air inlet end of the air blower communicates with the air inlet, the air outlet end of the air blower communicates with the air outlet, the baffle plate is fixedly arranged between the face cover and the intermediate cover, a second accommodating space is formed among the face cover, the intermediate cover and the baffle plate, and the air inlet communicates with the interior of the second accommodating space.

(8) Further, a filter screen is fixedly arranged at the air inlet.

(9) Further, the electric cutting assembly includes a power supply, a circuit control board, a motor and a cutter head assembly, the power supply, the motor and the air blower are electrically connected with the circuit control board, the motor drives the cutter head assembly to work, and the cutter head assembly is located at the air inlet end of the suction channel.

(10) Further, the electric cutting assembly includes a slide switch and a touch switch, and both the slide switch and the touch switch are electrically connected with the circuit control board.

(11) The invention has the following advantages and benefits in comparison with the conventional art: before the hair cutting and suction integrated electric clipper is used, the air inlet end of the external suction device is directly clamped at the air outlet end of the suction channel, so that the air inlet end of the external suction device communicates with the interior of the suction channel; when the external suction device works, air in the suction channel is extracted outside, so a negative pressure is formed in the suction channel; and after human hair or animal furs are cut by the electric cutting assembly, the clipped hair or furs are directly sucked in the suction channel and then sucked out by the external suction device, thereby preventing hair or furs from flying everywhere in the falling process. The hair cutting and suction integrated electric clipper is more favorable for collecting clipped hair or furs through the external suction devices.

(12) The above and other objects, features and advantages of the invention will become apparent from the following detailed description taken with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an electric clipper according to a first preferred embodiment of the invention;
- (2) FIG. 2 is a longitudinal sectional view of the electric clipper in FIG. 1;
- (3) FIG. 3 is an exploded view of the electric clipper in FIG. 1;
- (4) FIG. 4 is a longitudinal sectional view of an electric clipper according to a second preferred embodiment of the invention;
- (5) FIG. 5 is an exploded view of the electric clipper in FIG. 4; and
- (6) FIG. 6 is an exploded, perspective view of the electric clipper in FIG. 4.

DETAILED DESCRIPTION OF THE INVENTION

- (7) Referring to FIGS. 1 to 3, an electric clipper in accordance with a first preferred embodiment of the invention comprises a face cover 1, an intermediate cover 2, a bottom cover 3, an electric cutting assembly 4, and an external suction connector 5 as discussed in detail below.
- (8) The face cover 1 and the bottom cover 3 are buckled on two sides of the intermediate cover 2 respectively, a first accommodating space is formed between the intermediate cover 2 and the bottom cover 3, the electric cutting assembly 4 is fixedly arranged in the first accommodating space, a suction channel 11 is formed between the face cover 1 and the intermediate cover 2, and the an air inlet end 111 of the suction channel 11 is matched with the electric cutting assembly 4.
- (9) As shown in FIG. 1 specifically, the left end of the suction channel 11 is the air inlet end 111 of the suction channel 11, the right end of the suction channel 11 is an air outlet end 112 of the suction channel 11, and the electric cutting assembly 4 is arranged at the position close to the air inlet end 111 of the suction channel 11.
- (10) Two ends of the external suction connector penetrate through the external suction connector 5, i.e. the left end and the right end of the external suction connector 5 are penetrated, and one end of the external suction connector 5 is buckled at the air outlet end 112 of the suction channel 11, so as to communicate with the interior of the air suction channel 11.
- (11) The electric cutting assembly 4 includes a power supply 41, a circuit control board 42, a motor 43, a cutter head assembly 44 and a slide switch 45, the power supply 41, the motor 43, the air blower 46 and the slide switch 45 are electrically connected with the circuit control board 42, the motor 43 drives the cutter head assembly 44 to work, and the cutter head assembly 44 is located at the air inlet end 111 of the suction channel 11.
- (12) When the slide switch 45 is operated, the slide switch 45 sends a signal to the circuit control board 42, and the circuit control board 42 controls the motor 43 to work.
- (13) Working principle: the air inlet end of the external suction device is connected with the external suction connector 5, so that the air inlet end of the external suction device communicates with the interior of the suction channel 11; when the external suction device works, air in the suction channel 11 is extracted outside, so a negative pressure is formed in the suction channel 11; when the slide switch 45 is operated, the slide switch 45 sends a signal to the circuit control board 42, the circuit control board 42 controls the motor 43 to work, the motor 43 drives the cutter head assembly 44 to work; and after human hair or animal furs are cut, and the clipped hair or furs are directly sucked in the suction channel 11 and then sucked out by the external suction device.
- (14) Referring to FIGS. 4 to 6, an electric clipper in accordance with a second preferred embodiment of the invention is shown. The characteristics of the second preferred embodiment are substantially the same as that of the first preferred embodiment except the following:
- (15) The electric clipper comprises a face cover 1, an intermediate cover 2, a bottom cover 3, an electric cutting assembly 4, an air blower 6, and a baffle plate 7.
- (16) The face cover 1 and the bottom cover 3 are buckled on two sides of the intermediate cover 2

respectively, a first accommodating space is formed between the intermediate cover **2** and the bottom cover **3**, the electric cutting assembly **4** is fixedly arranged in the first accommodating space, a suction channel **11** is formed between the face cover **1** and the intermediate cover **2**, and the air inlet end of the suction channel **11** is matched with the electric cutting assembly **4**.

(17) As shown in FIG. **4** specifically, the left end of the suction channel **11** is an air inlet end **111** of the suction channel **11**, the right end of the suction channel **11** is an air outlet end **112** of the suction channel **11**, and the electric cutting assembly **4** is arranged at the position close to the air inlet end **111** of the suction channel **11**.

(18) An air inlet **21** is formed in the center of the intermediate cover **2**, an air outlet **31** is formed in the bottom cover **3**, the air blower **6** is fixedly arranged in the first accommodating space, the air inlet end of the air blower **6** communicates with the air inlet **21**, the air outlet end of the air blower **6** communicates with the air outlet **31**, the baffle plate **7** is fixedly arranged between the face cover **1** and the intermediate cover **2**, and a second accommodating space **12** is formed among the face cover **1**, the intermediate cover **2** and the baffle plate **7**.

(19) As shown in FIG. **4** specifically, it can also be understood in this way: the second accommodating space **12** is the left half part of the suction channel **11**.

(20) The air inlet **21** communicates with the interior of the second accommodating space **12**, and a filter screen is fixedly arranged at the air inlet **21**.

(21) When the air blower **6** works, air in the second accommodating space **12** is sucked in the air blower **6** through the air inlet **21** and then sucked out of the air outlet **31**, and the hairs or furs can be effectively blocked in the second accommodating space **12** by the filter screen.

(22) The electric cutting assembly **4** includes a power supply **41**, a circuit control board **42**, a motor **43**, a cutter head assembly **44**, a slide switch **45** and a touch switch **46**, the power supply **41**, the motor **43**, the air blower **6**, the slide switch **45** and the touch switch **46** are electrically connected with the circuit control board **42**, the motor **43** drives the cutter head assembly **44** to work, and the cutter head assembly **44** is located at the air inlet end **111** of the suction channel **11**.

(23) When the slide switch **45** is operated, the slide switch **45** sends a signal to the circuit control board **42**, and the circuit control board **42** controls the motor **43** to work; and when the touch switch **46** is operated, the touch switch **46** sends a signal to the circuit control board **42**, and the circuit control board **42** controls the air blower **6** to work.

(24) Working principle: during working, when the touch switch **46** is operated, the touch switch **46** sends a signal to the circuit control board **42**, the circuit control board **42** controls the air blower **6** to work, and the air blower **6** sucks air in the second accommodating space **12** from the air inlet **21** and then sucks the air out of the air outlet **31**, so a negative pressure is formed in the suction channel **11**; when the slide switch **45** is operated, the slide switch **45** sends a signal to the circuit control board **42**, the circuit control board **42** controls the motor **43** to work, the motor **43** drives the cutter head assembly **44** to work; and after human hairs or animal furs are cut, the clipped hair or furs are directly sucked in the second accommodating space **12**, and the hairs or furs can be effectively blocked in the second accommodating space **12** by the filter screen.

(25) While the invention has been described in terms of preferred embodiments, those skilled in the art will recognize that the invention can be practiced with modifications within the spirit and scope of the appended claims.

Claims

1. An electric clipper, comprising: a face cover; a bottom cover wherein the face cover and the bottom cover are buckled on two sides of an intermediate cover respectively; a first accommodating space formed between the intermediate cover and the bottom cover; an electric cutting assembly fixedly arranged in the first accommodating space; a suction channel formed between the face cover and the intermediate cover wherein an air inlet end of the suction channel is

matched with the electric cutting assembly; a baffle plate fixedly arranged between the face cover and the intermediate cover; an air blower fixedly arranged in the first accommodating space wherein an air inlet is formed in the center of the intermediate cover, an air outlet is formed in the bottom cover, an air inlet end of the air blower communicates with the air inlet, and an air outlet end of the air blower communicates with the air outlet; and a second accommodating space formed among the face cover, the intermediate cover, and the baffle plate wherein the air inlet communicates with the interior of the second accommodating space.

2. The electric clipper of claim 1, wherein the electric cutting assembly comprises a power supply, a circuit control board, a motor, and a cutter head assembly wherein the power supply, the motor, and the air blower are all electrically connected to the circuit control board, the motor drives the cutter head assembly to work, and the cutter head assembly is located at the air inlet end of the suction channel.

3. The electric clipper of claim 2, wherein the electric cutting assembly further comprises a slide switch electrically connected to the circuit control board, and a touch switch electrically connected to the circuit control board.
